

Applications of Artificial Neural Network

Assignment 3: Time-series Forecast and Transfer Learning Analysis

Experimental Report - LSTM with Current-Only Source-Domain Pretraining

Target Domain: Real-world fluid/pulp production line data

Source Domain: CNC milling current-feature dataset

Model: LSTM sequence forecaster

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Executive Summary

This report presents a time-series forecasting and cross-domain transfer learning experiment for HW3. The target-domain task is to predict the product-quality variable `Target_Value` in a real-world fluid/pulp production line. Following the assignment requirements, the model uses a 90-step historical lookback window and predicts a 60-step future horizon, while evaluation is performed specifically on the 60th future step.

The final experimental design uses an LSTM model because the problem is sequential: process variables such as current, pressure, flow, and consistency evolve over time before affecting product quality. For transfer learning, the source-domain data is derived from CNC milling, but only current-related source features are used. This current-only selection was chosen after source-feature subset testing showed that current features transferred better than the full CNC feature table.

The best model is TL Current-Only with the first LSTM layer frozen. It achieves $MAE = 14.8143$, $MSE = 484.8056$, $MAPE = 4.0394\%$, and $R2 = 0.7735$. This improves the scratch baseline by approximately 21.82% in MAE, 25.67% in MSE, 19.55% in MAPE, and +0.0782 in R2. Since the MAPE is far below 10%, the model satisfies the highest accuracy level in the HW3 grading rubric.

1. Assignment Requirements and Experimental Design

HW3 requires two major comparisons: (1) an ANN model trained from scratch on the target-domain industrial time-series data, and (2) a cross-domain transfer learning model pretrained on a related source-domain industrial dataset and fine-tuned on the

target domain. The assignment also requires sliding-window preprocessing with lookback = 90 and forecast horizon = 60, 60th-step evaluation using MAE, MSE, MAPE, and R2, prediction-vs-actual visualization, freezing-layer ablation, convergence analysis, and a cross-domain universality discussion.

1.1 Requirement Coverage

HW3 Requirement	How This Experiment Satisfies It
Task 1	
Model architecture rationality	Use LSTM because the task is sequential and target quality depends on delayed process dynamics.
Data preprocessing and windowing	Target labels are converted to numeric, invalid labels are removed, features are normalized, categorical material composition is one-hot encoded, and windows are generated only inside continuous chunks.
60th-step evaluation metrics	The model outputs a 60-step vector; metrics are computed using <code>y_pred[:, 59]</code> and <code>y_true[:, 59]</code> .
Visualization: Prediction plots	Prediction-vs-actual line/scatter plots are generated for each strategy.
Task 2	
Transfer dataset explanation	CNC current features are selected because both CNC and pulp/fluid production use current signals as load/operation-state indicators.
Layer freezing strategies	Scratch, full fine-tune, freeze-all-feature-layers, and freeze-first-LSTM-layer are compared.
Cross-domain universality	The report discusses why current/load signals may have transferable physical meaning across industrial domains.
Convergence analysis	Validation loss curves are compared across scratch and transfer strategies.

2. Dataset Description

2.1 Target-domain Dataset

The target domain consists of Anonymized_Train_Data.csv for training and Anonymized_Inference_Data.csv for testing. The data is a real-world continuous production-line dataset from Taiwan. Each row contains machine sensor readings and the target quality variable Target_Value. The original target data includes invalid non-numeric values such as Bad, so these values must be handled before supervised learning.

Item	Value
Training rows before target cleaning	71,401
Training rows after target cleaning	61,594
Training continuous chunks	134
Training windows	46,782
Testing rows before target cleaning	17,851
Testing rows after target cleaning	14,245
Testing continuous chunks	59
Testing windows	9,500
Feature dimension after encoding	119
Lookback / Horizon	90 / 60

2.2 Source-domain Dataset

The source domain is the CNC milling dataset ([Link](#), by “Grzegorz Piecuch” from Kaggle) represented by FeatureAndMetadata_Milling.csv. To find out the best features to be included for the source-domain dataset, I had done some experiments using different inclusion combinations (Current-only, full features, no min/max). These experiments showed that using all source features was not optimal because vibration/accelerometer features were too machining-specific. Therefore, the final report uses **current-only source features**. This source selection is meaningful because current signals represent machine load, resistance, and operation state, which are also important in the target fluid/pulp production process.

Source-domain Item	Value
Source feature variant	current_only
Source rows	968
Number of tools	14
Total source windows	712
Source train windows	569
Source validation windows	143

Selected current features	72
Source feature dimension	72
Max rows per tool	149
Median rows per tool	55.5

3. Model Architecture

The model is an **LSTM-based sequence forecaster**. LSTM is selected instead of a simple feedforward BPNN because the target problem is not a static tabular prediction problem. The product-quality value at the 60th future step **can depend on recent process trends and delayed effects from currents, pressures, flows, and other operating conditions**. LSTM can maintain hidden states over the 90-step input sequence and therefore is **more suitable for this time-series forecasting task**.

The target-domain model receives input with shape (batch_size, 90, feature_dim) and outputs a vector with shape (batch_size, 60). The final evaluation extracts the last element of this 60-step vector. The source-domain model uses a similar LSTM encoder but predicts CycleToFailureNormalized from source-domain current-feature sequences. Transfer learning is implemented by transferring the LSTM encoder and compatible hidden layers from the source model to the target model.

- Input projection layer: maps raw target/source features into the LSTM hidden dimension.
- LSTM encoder: learns temporal dependencies from the input sequence.
- Prediction head: maps the final LSTM state to the required output horizon.
- Loss function: mean squared error on the scaled target values.
- Optimizer: Adam with early stopping based on validation loss.

4. Data Preprocessing and Windowing

The preprocessing follows the HW3 hint logic. First, Target_Value is converted to numeric using error coercion. Rows where Target_Value cannot be converted are removed. Second, the data is sorted by Sequence_ID and Timestamp. Third, time gaps are calculated, and a new continuous chunk is created whenever the gap is greater than 5 minutes or the Sequence_ID changes. This is important because the assignment strictly prohibits windowing across production-line shutdowns or network-disconnection gaps.

After data cleaning and chunking, sliding windows are generated inside each continuous chunk only. Each input sample X contains 90 historical time steps. The target y contains the next 60 future Target_Value steps. During evaluation, only the final step of this 60-step horizon is used. Therefore, the exact evaluation target is $y[:, 59]$, corresponding to the 60th future step.

Numerical features are standardized with StandardScaler. The categorical variable Material_Composition is one-hot encoded with handle_unknown='ignore' so that unseen material combinations in the inference set do not crash the test pipeline.

5. Transfer Learning Design

The transfer-learning experiment uses current-only CNC features as the source domain. This choice is based on the idea that current signals are physically meaningful in both domains. In CNC machining, current reflects spindle and axis load under cutting resistance. In pulp/fluid production, motor current also reflects load, resistance, and process operating state. Therefore, current features are more likely to transfer than vibration features, which are specific to CNC machining and do not directly exist in the target-domain sensor set.

Four strategies are compared: scratch training, current-only full fine-tuning, current-only transfer with all feature layers frozen, and current-only transfer with the first LSTM layer frozen. The freezing ablation studies how much of the source-domain representation should be reused directly and how much should be adapted to the target domain.

6. Experimental Results

Experiment	MAE	MSE	MAPE (%)	R2	Best Epoch	Trainable Params	Total Params
TL Current-Only Freeze First LSTM Layer	14.8143	484.8056	4.0394	0.7735	1	45,020	78,300
TL Current-Only Freeze All Feature Layers	18.1008	632.9048	4.8528	0.7043	6	9,660	78,300
Scratch	18.9484	652.2506	5.0211	0.6952	4	78,300	78,300

TL Current-Only Full Fine-tune	18.3900	607.3049	5.0329	0.7162	1	78,300	78,300
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The best-performing model is **TL Current-Only Freeze First LSTM Layer**, with MAE = 14.8143, MSE = 484.8056, MAPE = 4.0394%, and R2 = 0.7735. Compared with the scratch baseline, this reduces MAE by 21.82%, MSE by 25.67%, and MAPE by 19.55%, while increasing R2 by 0.0782.

All models achieve MAPE below 10%, but the partial-freezing transfer model is the strongest. This indicates that source-domain current features provide useful low-level temporal representations, while the target-domain model still needs substantial adaptation in higher layers.

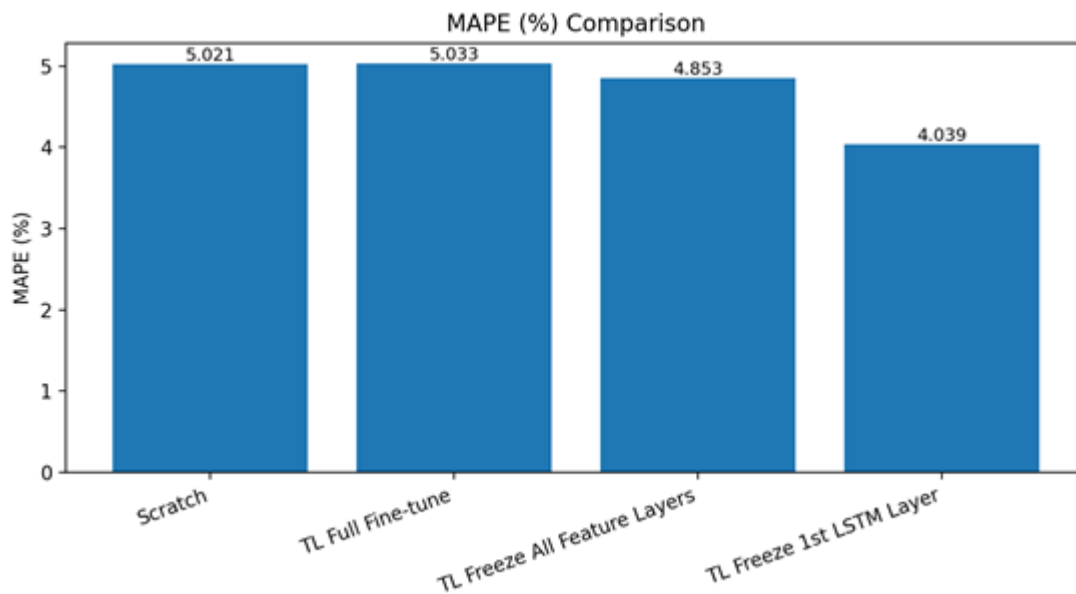


Figure 1. MAPE comparison across scratch and current-only transfer strategies.

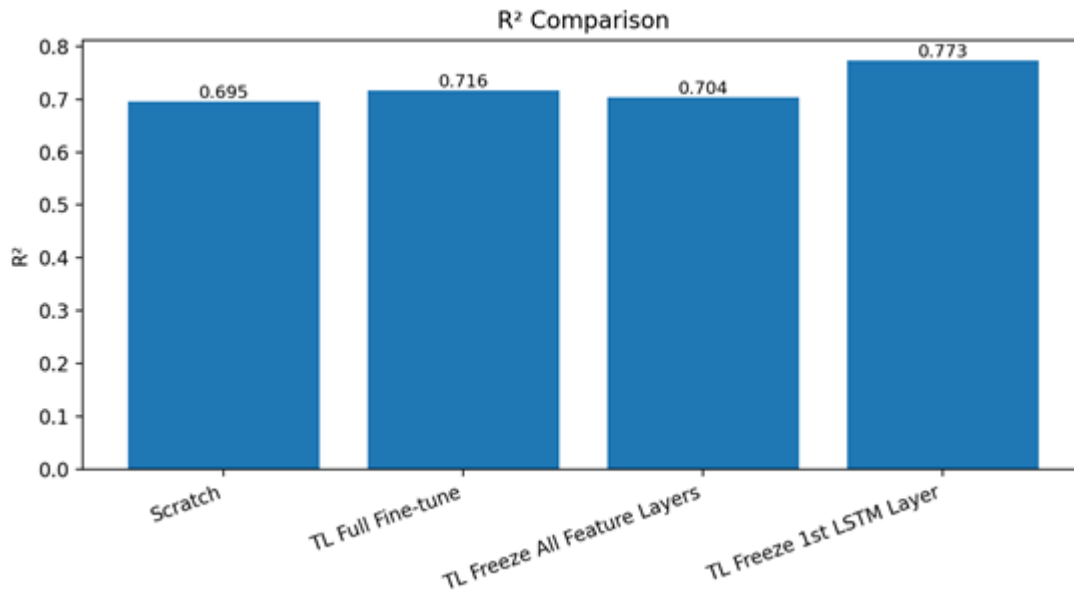


Figure 2. R² comparison across scratch and current-only transfer strategies.

7. Prediction Plot Analysis

The prediction plots compare the blue predicted line against the red actual target values at the 60th forecast step. All models capture the main process trend, especially the drop around the middle region and later recovery. However, there is still a visible early segment where actual values stay near 200. This likely corresponds to special operating conditions or remaining suspicious target/sensor values. I had tried to find potential anomalies (e.g. 0 as a value for the sensor data) and remove the suspicious entries. This seems to be cleaner with all the 200 values gone, but this seems to be way too aggressive in the clearing method, and yields a very bad output (R² becomes negative, meaning the predictions are worse than mean).

The best transfer model shows better tracking of the target trend than scratch training, especially after the major drop around sample index 4000. It also reduces the overall error while keeping the prediction line smoother than the actual scatter points. This smoothing is expected because the model learns average sequence dynamics and may not fully capture sudden spikes.

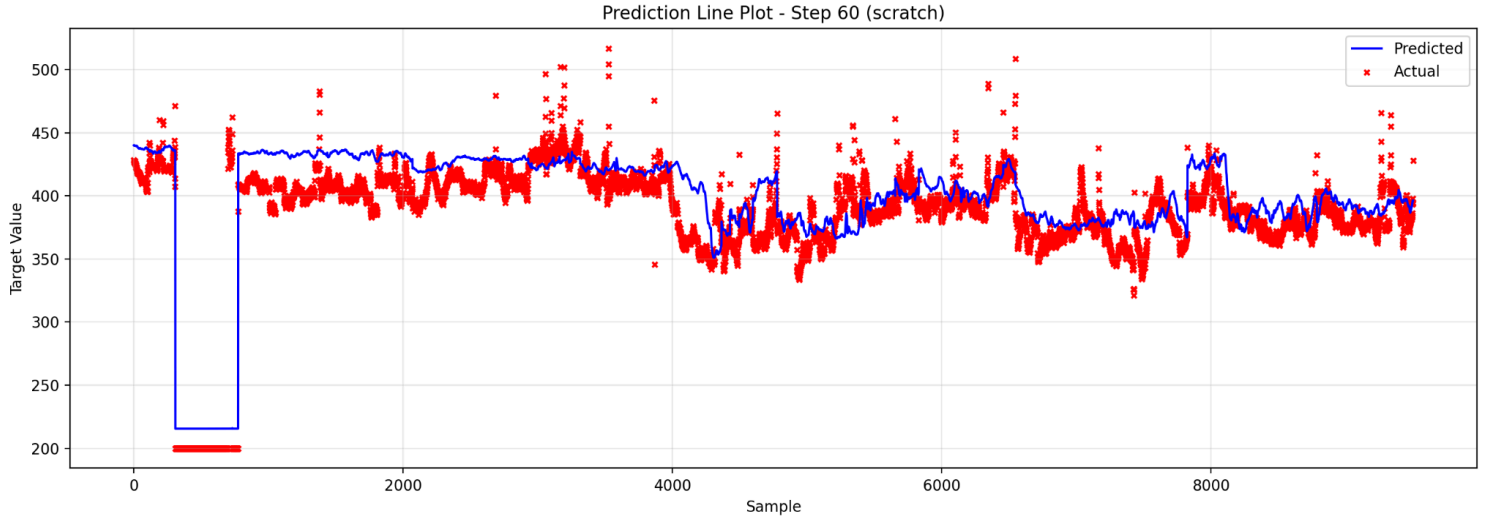


Figure 3. Prediction-vs-actual plot for scratch training at the 60th step.

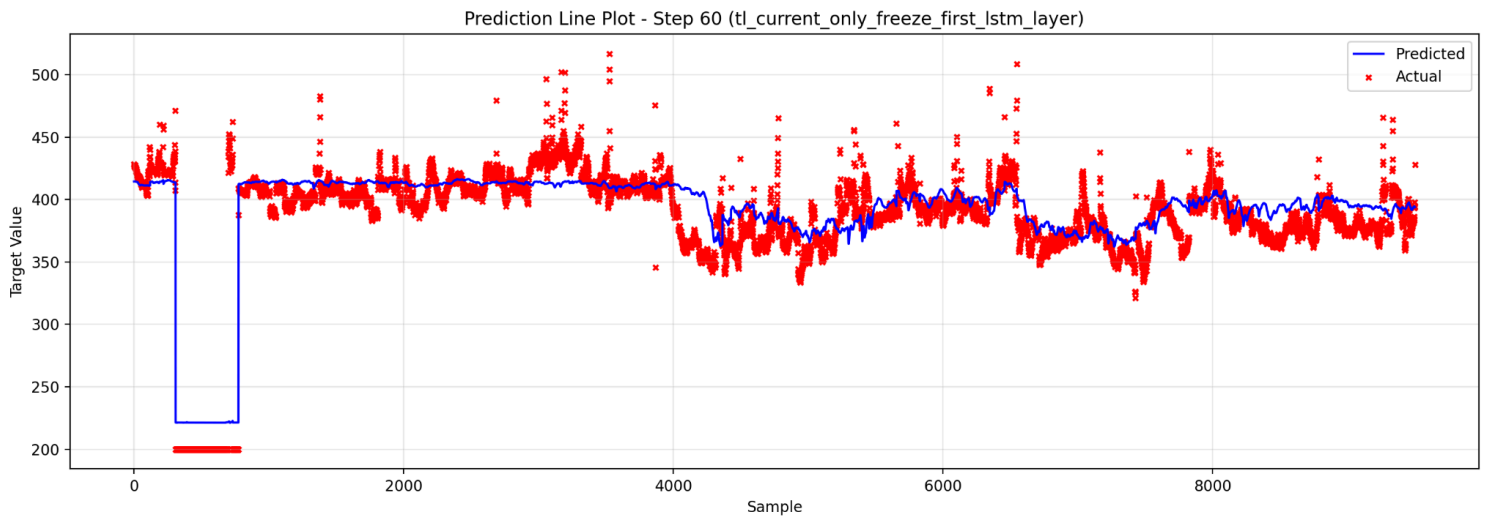


Figure 4. Prediction-vs-actual plot for the best transfer model: current-only source with first LSTM layer frozen.

08. Interpretation of Transfer Data and Freezing Strategies

The final transfer data was selected from the CNC milling source domain, but only current-related columns were used. This choice was not arbitrary. The target-domain production line contains many current-related process variables, and current is a physically meaningful signal in industrial systems because it reflects motor load, process resistance, and changes in operating conditions. Therefore, current signals provide a stronger cross-domain bridge between CNC machining and fluid/pulp production than CNC accelerometer or vibration signals.

The original source-domain feature table contained a broad set of CNC features, including accelerometer min/max/mean/std/skew/kurtosis and current min/max/mean/std/skew/kurtosis. However, the target production-line dataset does not contain direct CNC vibration or accelerometer measurements. If the full CNC feature table is used, the pretrained model may learn machining-specific vibration behavior that does not correspond to the target domain. This can lead to negative transfer. By selecting only current-related source features, the source-domain representation becomes more aligned with the target process, because both domains share the physical meaning of current as a load and operation-state indicator.

The source-domain current-only dataset is also suitable for LSTM pretraining because the CNC feature table can be ordered by tool cycle, allowing each tool history to become a sequence. The source task is to predict CycleToFailureNormalized from current-feature sequences. Although this label is not the same as the target production-quality value, both tasks require the model to learn how machine operating signals change over time before an important outcome changes. Thus, the source data is useful mainly as a temporal representation initializer, not as a direct predictor of the target variable.

The freezing results support this interpretation. Freezing the first LSTM layer gives the best result, which suggests that the first recurrent layer learned useful low-level temporal patterns from source current features. These lower-level patterns may include local trend changes, machine-load transitions, and smoothness or instability characteristics that are physically meaningful across domains. The higher LSTM layer and prediction head, however, must remain trainable because the target-domain process and output definition are different from the CNC source-domain task.

Freezing all feature layers is weaker because it prevents the model from adapting enough to the target production process. Full fine-tuning is also not optimal in this run because allowing all transferred weights to update may overwrite useful source-domain temporal patterns too quickly. Therefore, the best trade-off is to preserve the most transferable low-level current/load representation in the first LSTM layer while fine-tuning the higher recurrent and output layers for the target production-quality prediction task.

09. Cross-domain Universality Analysis

The key cross-domain question is whether knowledge learned from CNC machining can help forecast quality in a fluid/pulp production line. The answer from this experiment is yes, but only when the source features are physically aligned with the target process. The final model uses only current-related CNC features because

current is a general industrial signal connected to motor load, mechanical resistance, and operation state.

In CNC machining, current changes when tool load, material resistance, and cutting condition change. In pulp/fluid production, current also changes when machines experience different load, flow resistance, or process demand. Therefore, although the industries are different, current signals can contain a shared physical meaning: they reflect how hard machines are working under changing process conditions.

This is why the current-only transfer performs better than using the full CNC feature table. The full source table includes accelerometer and vibration features that are important for CNC machining, but those features have weak correspondence to the target process. Including them can cause negative transfer because the model may learn source-specific vibration patterns that do not generalize to pulp/fluid production. Current-only pretraining removes much of this source-specific noise and focuses the transferred representation on a more universal physical signal.

10. Convergence Analysis

The convergence curve compares validation MSE loss across scratch training and transfer learning strategies. The best transfer strategy, freezing the first LSTM layer, starts with relatively low validation loss and reaches its best validation epoch very early. This suggests that the source pretrained current-based encoder provides a useful initialization. However, the full fine-tuning curve is less stable and does not outperform the partial-freezing strategy in final test metrics.

The freezing results reveal an important pattern. Freezing all feature layers reduces the number of trainable parameters drastically, but its performance is only slightly better than scratch. Full fine-tuning improves R^2 but does not improve MAPE compared with scratch. The first-layer-freezing setting performs best, suggesting that the earliest recurrent layer captures transferable current/load dynamics, while the higher LSTM layer and prediction head should remain adaptable to the target production process.

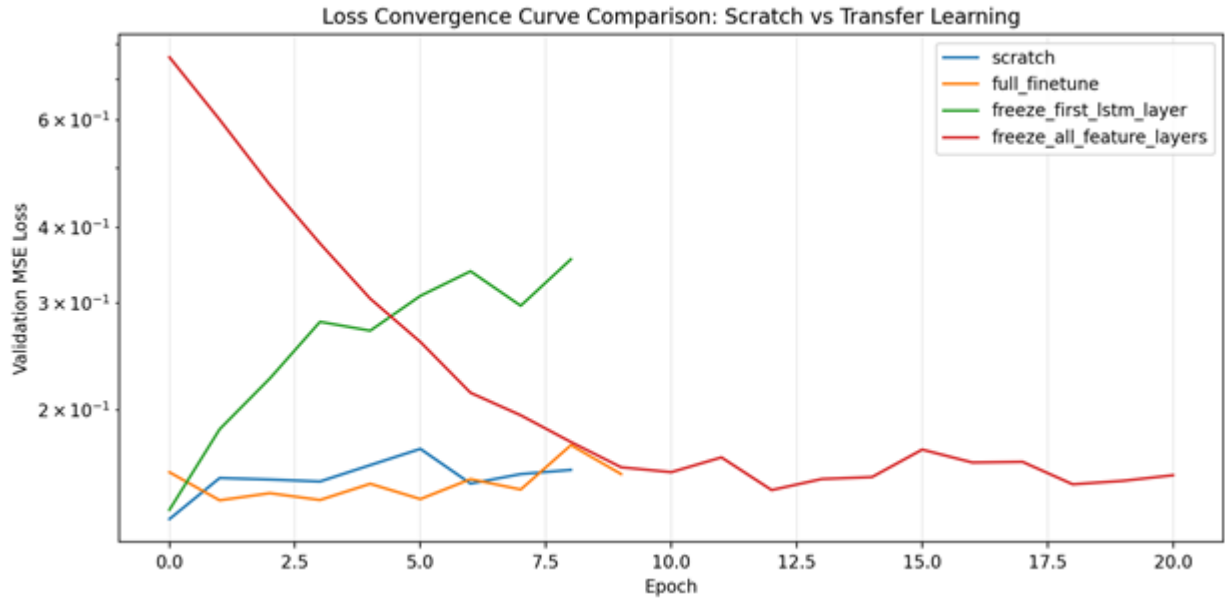


Figure 5. Loss convergence comparison: scratch versus current-only transfer learning.

11. Conclusion

This experiment successfully implements the HW3 time-series forecasting and transfer-learning requirements. The LSTM model predicts the 60th future step of Target_Value using a 90-step lookback window and reports MAE, MSE, MAPE, and R2. The best result is obtained by transferring from a current-only CNC source-domain model and freezing the first LSTM layer during target fine-tuning.

The best model achieves MAPE = 4.0394%, which is well below the 10% threshold in the grading rubric. The result demonstrates that cross-domain transfer learning is useful when the source features are selected based on physical similarity. The source dataset itself is not bad; rather, effective source-feature selection is essential. Current-related features provide transferable information because they represent machine load and operating state across both CNC machining and pulp/fluid production.